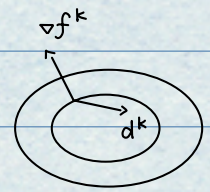
$\min f(x)$

$x_{k+1} = x_k + \alpha_k d_k$

Descent Direction d_k : $\langle d_k , \nabla f_k \rangle < 0$



Steepest Descent $d_k = -\nabla f_k$ $\qquad x_{k+1} = x_k - \alpha \nabla f_k$

Newton's Method $d_k = -H_k^{-1} \nabla f_k$ $\qquad x_{k+1} = x_k - \alpha H_k^{-1} \nabla f_k$

CG Method $d_k = -\nabla f_k + \beta_k d_{k-1}$ $\qquad \beta_k^{FR} = \dfrac{\|\nabla f_k\|^2}{\|\nabla f_{k-1}\|^2}$

Step size $\alpha_k = \operatorname{argmin} f(x_k + \alpha_k d_k)$

$= \langle \nabla f(x_k + \alpha_k d_k) , d_k \rangle = 0$ 的根

e.g. $f = \frac{1}{2} x^T A x - \langle x , b \rangle$

$\alpha_k = -\dfrac{\langle r_k , p_k \rangle}{\langle p_k , p_k \rangle_A}$. $\quad r_k = A x_k - b$

Backtracking α 减小 until $f_{k+1} - f_k \le C_1 \alpha \langle d_k , \nabla f_k \rangle$

Armijo conditio